NeuMathe 题库导出

共 6 道题目 | 2025-12-19 17:03

1.

设 $z=f\left(x,x^2y,3x-e^y\right)$，其中 f 有连续偏导数，求 $\frac{\partial z}{\partial x},\frac{\partial z}{\partial y}=()$

(A) $f_1'+2xyf_2'+3f_3',\quad x^2f_2'-e^yf_3'$ 【答案】

(B) $f_1'+2xyf_2'+3f_3',\quad x^2f_2'+e^yf_3'$

(C) $f_1'+xyf_2'+3f_3',\quad x^2f_2'-e^yf_3'$

(D) $f_1'+2xyf_2'+3f_3',\quad xyf_2'-e^yf_3'$

【解析】

解答设 $u=x^2y,v=3x-e^y$，则

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial u}\cdot 2xy+\frac{\partial f}{\partial v}\cdot 3=f_1'+2xyf_2'+3f_3'$$
$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial y}\cdot 0+\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}=\frac{\partial f}{\partial u}\cdot x^2-\frac{\partial f}{\partial v}\cdot e^y=x^2f_2'-e^yf_3'$$

2.

设函数 $u=\varphi[x+\psi(y)]$ ，则下列等式正确的是（）.

(A) $\frac{\partial u}{\partial x}\frac{\partial^2u}{\partial x\partial y}=\frac{\partial u}{\partial y}\frac{\partial^2u}{\partial y^2}$

(B) $\frac{\partial u}{\partial x}\frac{\partial^2u}{\partial x\partial y}=\frac{\partial u}{\partial y}\frac{\partial^2u}{\partial x\partial y}$

(C) $\frac{\partial u}{\partial x}\frac{\partial^2u}{\partial y^2}=\frac{\partial u}{\partial y}\frac{\partial u}{\partial x^2}$

(D) $\frac{\partial u}{\partial x}\frac{\partial^2u}{\partial x\partial y}=\frac{\partial u}{\partial y}\frac{\partial^2u}{\partial x^2}$ 【答案】

【解析】

由于$\frac{\partial u}{\partial x}=\varphi',\frac{\partial u}{\partial y}=\varphi'\psi',\frac{\partial^2u}{\partial x\partial y}=\varphi''\psi',\frac{\partial^2u}{\partial x^2}=\varphi''$

因此$\frac{\partial u}{\partial x}\frac{\partial^2u}{\partial x\partial y}=\varphi'\psi'\varphi''=\frac{\partial u}{\partial y}\frac{\partial^2u}{\partial x^2}$

3.

设 $z=f\left(x,\frac{x}{y}\right)$，则 $\frac{\partial^2 z}{\partial x^2}=()$

A $\frac{\partial^2 f}{\partial u^2}+\frac{\partial^2 f}{\partial v^2}+\frac{1}{2y^2}\frac{\partial^2 f}{\partial u\partial v}$

B $\frac{\partial^2 f}{\partial u^2}+\frac{1}{y}\frac{\partial^2 f}{\partial v^2}+\frac{1}{2y^2}\frac{\partial^2 f}{\partial u\partial v}$

C $\frac{\partial^2 f}{\partial u^2}+\frac{2}{y}\frac{\partial^2 f}{\partial v\partial u}+\frac{1}{y^2}\frac{\partial^2 f}{\partial v^2}$ 【答案】

D $\frac{\partial^2 f}{\partial u\partial v}+\frac{2}{y}\frac{\partial^2 f}{\partial v^2}+\frac{1}{2y^2}\frac{\partial^2 f}{\partial u^2}$

【解析】

$$\frac{\partial^2 z}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial u}+\frac{1}{y}\frac{\partial f}{\partial v}\right)=\frac{\partial^2 f}{\partial u^2}\frac{\partial u}{\partial x}+\frac{\partial^2 f}{\partial u\partial v}\frac{\partial v}{\partial x}+\frac{1}{y}\left(\frac{\partial^2 f}{\partial u\partial v}\frac{\partial u}{\partial x}+\frac{\partial^2 f}{\partial v^2}\frac{\partial v}{\partial x}\right)$$
$$=\frac{\partial^2 f}{\partial u^2}+\frac{2}{y}\frac{\partial^2 f}{\partial v\partial u}+\frac{1}{y^2}\frac{\partial^2 f}{\partial v^2}$$

4.

$u=f(x,xy)$ 具有二阶偏导数，求 $\frac{\partial^2 u}{\partial x\partial y}=()$

A $yf''_{12}+f'_2+xyf''_{22}$

B $xf''_{12}+f'_2+xyf''_{22}$ 【答案】

C $xyf''_{12}+f'_2+xf''_{22}$

D $xf''_{12}+xyf''_{22}$

【解析】

$\frac{\partial u}{\partial x}=f'_1+yf'_2$，$\frac{\partial^2 u}{\partial x\partial y}=xf''_{12}+f'_2+xyf''_{22}$

5.

设 $z=f(2x-y)+g(x,xy)$，其中函数 $f(t)$ 二阶可导，$g(u,v)$ 具有连续的二阶偏导数，则 $\frac{\partial^2 z}{\partial x\partial y}$ 等于（ ）

Ⓐ $-2f''+g''_{21}+g'_2+xyg''_{22}$

Ⓑ $-2f''+g''_{21}+g'_2+yg''_{22}$

Ⓒ $-2f''+xg''_{21}+g'_2+xyg''_{22}$ 【答案】

Ⓓ $-2f''+xg''_{21}+g'_2+yg''_{22}$

【解析】

由于混合偏导数在连续条件下与求导次序无关，可以先求 $\frac{\partial z}{\partial x}$，也可以先求 $\frac{\partial z}{\partial y}$。
方法一：先求 $\frac{\partial z}{\partial x}$，由复合函数求导法，

$$\frac{\partial z}{\partial x}=f'\frac{\partial}{\partial x}(2x-y)+g'_1\frac{\partial}{\partial x}(x)+g'_2\frac{\partial}{\partial x}(xy)=2f'+g'_1+yg'_2$$

再对 y 求偏导，得

$$\begin{aligned}\frac{\partial^2 z}{\partial x\partial y}=&\frac{\partial}{\partial y}\left(2f'+g'_1+yg'_2\right)=2f''\frac{\partial}{\partial y}(2x-y)\\&+\left[g''_{11}\frac{\partial}{\partial y}(x)+g''_{12}\frac{\partial}{\partial y}(xy)\right]+\left[g'_2+yg''_{21}\frac{\partial}{\partial y}(x)+yg''_{22}\frac{\partial}{\partial y}(xy)\right]\\=&-2f''+g''_{11}\cdot 0+xg''_{12}+g'_2+yg''_{21}\cdot 0+xyg''_{22}\\=&-2f''+xg''_{21}+g'_2+xyg''_{22}.\end{aligned}$$

方法二：先求 $\frac{\partial z}{\partial y}$，$\frac{\partial z}{\partial y}=f'\frac{\partial}{\partial y}(2x-y)+g'_1\frac{\partial}{\partial y}(x)+g'_2\frac{\partial}{\partial y}(xy)=-f'+xg'_2$，
再对 x 求偏导数，得 $\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial^2 z}{\partial y\partial x}=\frac{\partial}{\partial x}(-f'+xg'_2)$

$$\begin{aligned}&=-f''\frac{\partial}{\partial x}(2x-y)+g'_2+xg''_{21}\frac{\partial}{\partial x}(x)+xg''_{22}\frac{\partial}{\partial x}(xy)\\&=-2f''+g'_2+xg''_{21}+xyg''_{22}.\end{aligned}$$

【相关知识点】复合函数求导法则：若 $u=u(x,y)$ 和 $v=v(x,y)$ 在点 (x,y) 处偏导数存在，函数 $z=f(u,v)$ 在对应点 (u,v) 具有连续偏导数，则复合函数 $z=f[u(x,y),v(x,y)]$ 在点 (x,y) 处的偏导数存在，且

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x},\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}$$

6.

设 $z=xf\left(2x,y^2/x\right),f$ 具有连续的二阶偏导数，求 $\frac{\partial^2 z}{\partial x\partial y}=$（　）

A $2y\frac{\partial^2 f}{\partial u\partial v}-\frac{2y^3}{x^2}\frac{\partial^2 f}{\partial v^2}$

B $4y\frac{\partial^2 f}{\partial u\partial v}-\frac{2y^3}{x^2}\frac{\partial^2 f}{\partial v^2}$

【答案】

C $4y\frac{\partial^2 f}{\partial u\partial v}-\frac{2x^3}{y^2}\frac{\partial^2 f}{\partial v^2}$

D $2y\frac{\partial^2 f}{\partial u\partial v}-\frac{2x^3}{y^2}\frac{\partial^2 f}{\partial v^2}$

【解析】

令$u=2x,v=y^2/x,$ 则$z=xf(u,v).$

$$\frac{\partial z}{\partial y}=x\left(\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}\right)=x\left(\frac{\partial f}{\partial u}\cdot 0+\frac{\partial f}{\partial v}\cdot\frac{2y}{x}\right)=2y\frac{\partial f}{\partial v}$$

$$\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial}{\partial x}\left(2y\frac{\partial f}{\partial v}\right)=2y\left(\frac{\partial^2 f}{\partial u\partial v}\frac{\partial u}{\partial x}+\frac{\partial^2 f}{\partial v^2}\frac{\partial v}{\partial x}\right)$$

$$=2y\left[\frac{\partial^2 f}{\partial u\partial v}\cdot 2+\frac{\partial^2 f}{\partial v^2}\cdot\left(-\frac{y^2}{x^2}\right)\right]$$

$$=4y\frac{\partial^2 f}{\partial u\partial v}-\frac{2y^3}{x^2}\frac{\partial^2 f}{\partial v^2}$$